

Unit 2: Technical Sketching for Aerospace Design

Quiz 2A: Concept Sketching, Annotations, and Design Intent

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- What is the main purpose of a concept sketch?
 - To be perfect artwork
 - To communicate an idea quickly and clearly
 - To replace all testing
 - To decorate the notebook
- What does an annotation do?
 - Names only the title
 - Explains function, material, connection, or design intent
 - Erases mistakes
 - Sets the grade automatically
- Which is the best annotation?
 - Foam here
 - Foam padding protects the sensor during landing impact
 - Thing
 - Nice part
- Design intent means...
 - The reason behind a design choice
 - The color of a marker
 - The final page count
 - The computer used
- A useful concept sketch should include...
 - Labels and notes explaining major features
 - Only shading
 - No text
 - Only dimensions
- Why are rough sketches valuable?
 - They help explore ideas before final drawings
 - They are always final answers
 - They remove the need to think
 - They replace CAD
- Which sketch feature improves communication?
 - Unlabeled arrows
 - Clear labels and notes
 - Hidden notes
 - Tiny unreadable text

8. What should students do if someone misunderstands their sketch?
- A. Ignore them
 - B. Improve labels, views, or annotations
 - C. Start over silently
 - D. Delete all notes

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|-----------------------------|---|
| _____ Concept sketch | A. A quick drawing used to communicate an early idea. |
| _____ Annotation | B. A note explaining how or why a feature works. |
| _____ Design intent | C. The reason behind a design choice. |
| _____ Label | D. A short name identifying a part or feature. |
| _____ Sketch clarity | E. How easily another person can understand a sketch. |

Part C: Short Response

1. Explain the difference between a label and an annotation.

2. Describe how annotations can improve an aerospace concept sketch.

Unit 2: Technical Sketching for Aerospace Design

Quiz 2B: Isometric Sketching and Pictorial Views

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- An isometric sketch shows...
 - Only the front view
 - Only a top view
 - A 3D-looking view using equal angled axes
 - Only hidden lines
- Why is isometric paper useful?
 - It provides angled guide lines for 3D sketches
 - It makes every sketch automatic
 - It replaces dimensions
 - It prevents mistakes completely
- Which object is best communicated with an isometric sketch?
 - A single written sentence
 - A 3D bracket or payload housing
 - A class roster
 - A quiz score
- In an isometric sketch, parallel edges of a rectangular object usually...
 - Remain parallel in the sketch
 - Always curve
 - Disappear
 - Become vertical only
- What should an isometric sketch communicate?
 - Overall form and major features
 - Only final cost
 - Only material source
 - Only due date
- Which practice improves isometric sketches?
 - Using guide axes and keeping proportions consistent
 - Random angles
 - No construction lines
 - Drawing only from memory
- Why might aerospace engineers use pictorial sketches?
 - To communicate shape before modeling or fabrication
 - To avoid documentation
 - To replace all testing
 - To decorate slides

8. A common isometric sketch error is...
- A. Showing three visible faces
 - B. Changing edge angles randomly
 - C. Using labels
 - D. Keeping proportions

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|--------------------------------|---|
| _____ Isometric sketch | A. A 3D-style sketch using angled axes. |
| _____ Pictorial view | B. A drawing that shows an object in a realistic 3D-like way. |
| _____ Construction line | C. A light guide line used to help build a sketch. |
| _____ Visible edge | D. An edge that can be seen from the chosen view. |
| _____ Proportion | E. The relative size relationship between features. |

Part C: Short Response

1. Explain why isometric graph paper is helpful for sketching 3D parts.

2. Describe one situation where an isometric sketch would be more useful than a single front view.

Unit 2: Technical Sketching for Aerospace Design

Quiz 2C: Multiview Drawings, Line Conventions, and Dimensioning

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. A multiview drawing usually shows...
A. Several 2D views of the same object
B. Only one 3D view
C. Only a picture from the internet
D. Only a title block
2. Which views are commonly used in a basic multiview drawing?
A. Front, top, and right side
B. Left diagonal only
C. Back only
D. None
3. Hidden lines show...
A. Edges not visible from that view
B. The page border
C. Final answers
D. Text labels
4. Centerlines are often used to show...
A. Centers of holes or symmetrical features
B. Random decoration
C. Due dates
D. Material cost
5. Dimension lines show...
A. The measured size or distance of a feature
B. Only color
C. Only the title
D. Only hidden edges
6. Extension lines connect...
A. A feature to its dimension
B. A student to a grade
C. Two unrelated sketches
D. The title to the footer
7. Good dimensioning should be...
A. Clear, complete, and not duplicated unnecessarily
B. Random and repeated everywhere
C. Only verbal
D. Hidden from the viewer

8. Why are line conventions important?
- A. They create a shared visual language for technical drawings B. They make drawings artistic only
- C. They remove the need for accuracy D. They are optional in engineering

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|--------------------------------|---|
| _____ Multiview drawing | A. A set of 2D views used to fully describe a 3D object. |
| _____ Hidden line | B. A dashed line showing an edge not visible from that view. |
| _____ Centerline | C. A line showing the center of a hole, circle, or symmetrical feature. |
| _____ Dimension line | D. A line that displays the measurement value. |
| _____ Extension line | E. A line that extends from a feature to a dimension line. |

Part C: Short Response

1. Explain why multiview drawings are useful for manufacturing parts.

2. Describe one mistake that can make a dimensioned drawing difficult to use.
